

KSHITIJ DURAPHE

Boston, MA | kshitijduraphe5@gmail.com | [LinkedIn](#) | [Github](#) | [Portfolio](#) | 314-886-3066

PROFESSIONAL EXPERIENCE

Thespian Labs

Somerville, MA

AI Engineer

Nov 2025 – Feb 2026

- Managed the lifecycle of custom **Text2Motion** deep learning models from development, deployment, and monitoring, ensuring optimal performance and stability on **GCP**. Published an open-source library for MLOps: [Link](#).
- Developed and maintained batch processing ETL pipelines for data processing of **5000+** hours of human performance data.
- Collaborated with **cross-functional teams** to build a SOTA **foundation model** for **controllable digital human performance generation** on the frontier of **human-computer interaction**.

Absentia Technologies

Boston, MA

Founding Machine Learning Engineer

Jan 2025 – Nov 2025

- Architected and deployed a production-ready SaaS video analysis platform from the ground up on AWS, establishing core AI service infrastructure and a full CI/CD pipeline with Terraform and Docker, reducing deployment cycles from days to **<2 hours**.
- Built the core infrastructure and APIs enabling a fleet of autonomous **AI agents** to **read** video streams, **understand** complex events through a reasoning engine, and **act** by flagging anomalies in real-time using Python, PyTorch, and Kafka on Kubernetes (K8S).
- Designed and implemented a “ChatGPT for videos” pipeline using frame-level SVM classification to route frames to offline ViM/SwinV2 enhancement, with precomputed AI agent answers delivering **<1s p99** latency for common queries and **30s** for novel questions, enabling efficient interactive video Q&A at scale
- Established a rigorous, automated **evaluation pipeline** to benchmark agent performance and API latency, reducing model regressions by 40%; improved system stability to **99.9%** by resolving a critical memory leak during on-call duties.
- Developed a multimodal Video Question-Answering (VideoQA) system for complex temporal reasoning, improving answer accuracy by **25%**, and built a synthetic data pipeline with diffusion models to reduce false positives by **15%**.

The KeelWorks Foundation

Oak Harbor, WA

Software Engineer (Machine Learning Applications)

July 2024 – Jan 2025

- Engineered a production-scale **retrieval and reasoning system** using a RAG pipeline in TypeScript/Python with LangChain, delivering a search API with **<5s p95** latency and **92% context relevance (MRR)** across 2,500+ documents on a PostgreSQL backend.
- Executed an aggressive model optimization strategy for production deployment, reducing model size by **50%** and increasing inference speed by **80%** using **8-bit GPTQ quantization** and knowledge distillation while maintaining **>90%** task accuracy.
- Pioneered a synthetic data generation workflow by fine-tuning **Mistral-7B**, expanding the training dataset for downstream tasks by **30%** and significantly improving model robustness; deployed all services via a Dockerized GitLab CI/CD workflow.

Space Physics Lab, Boston University

Boston, MA

Graduate Research Assistant (ML Applications)

Oct 2022 – May 2024

- Led R&D of a multimodal system to parse and reason over noisy, unstructured sensor data; Architected a high-throughput, distributed data ingestion system using Kafka, Dask, and AWS S3, slashing processing time for 3TB+ datasets from **>24 hours** to **<3 hours**.
- Developed a low-latency forecasting system achieving **<80ms p90** latency at **25 predictions/sec** by implementing async processing and request batching, which cut initial latency by 40%.
- Implemented **generative inpainting** and **SwinIR super-resolution** pipelines to reconstruct corrupted sensor data, improving data quality and signal-to-noise ratio for downstream predictive models by over 60%.

PROJECTS

ArkOS (MIT)

Jun 2025 – Present

- DevOps, documentation (**Mintlify**) and general development (frontend/backend) for ArkOS, an open source interface for a local LLM agent building utilizing long term memory for personalized requests

Halo AI (Stealth startup incubated at Columbia University)

Dec 2023 – Aug 2024

- Developed and deployed a federated learning pipeline and ensemble of LLMs for on-device **agentic conversational assistants**, cutting inference latency by **27%** to **<1.5s**.
- Architected a scalable MLOps pipeline on AWS (**EC2/S3/Lambda, SageMaker**), automating CI/CD for federated learning models and reducing data preparation time by **70%**.

- Implemented an **evaluation pipeline** to monitor model performance, utilizing knowledge distillation and 8-bit quantization to improve inference speed by **80%** while preserving **90%** accuracy.

BeatQraft (MIT iQuHACK 2023 Hackathon)

Jan 2023

- Built a **distributed quantum generative AI** service at the MIT iQuHACK-23 hackathon, placing **2nd** out of 1000+ teams. [Link](#)

TECHNICAL SKILLS

Programming & Databases:	Python, C++, TypeScript, SQL (Postgres), MongoDB, Redis
MLOps & Cloud Platforms:	Docker, AWS, GCP, Kafka, Dask, Terraform, Kubernetes, Modal, MLflow
Machine Learning & Tools:	PyTorch, GenAI, RAG, LLMs, Computer Vision, Retrieval Systems, Distributed Systems, AI Code Assistants
Core Concepts:	Software Architecture, Product Intuition, High-Performance Computing, Scalability, Agile

EDUCATION

Boston University

Boston, MA

Master of Science with Thesis in Electrical and Computer Engineering
GPA: 3.8/4

Sep 2022 – May 2024

College of Engineering Pune

Pune, India

Bachelor of Technology in Electrical Engineering, Minor in CS
GPA: 3.83/4

Aug 2018 – June 2022

PUBLICATIONS

1. **Optimizing Solar Panel Tilt using Machine Learning Techniques**, [GPECOM 2021](#).
Proposes an XGBoost-based approach to maximize energy generation from solar plants.
2. **The Platonic Universe: Do Foundation Models See the Same Sky?**, [NeurIPS ML4PS 2025 - Spotlight Paper](#).
Investigates if different foundation models see the same underlying astrophysical phenomena and develops custom foundation models to better learn underlying astrophysics.